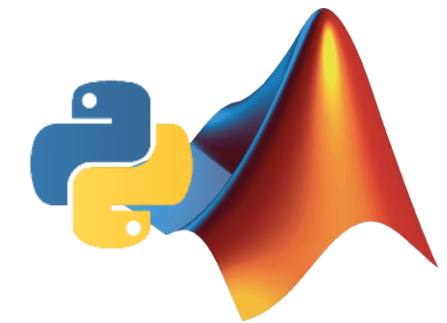


MATLAB & Python: Supercharge Your Projects



Kantika Wongkasem
Application Engineer
Ascendas Systems Co.,Ltd



“MATLAB & Python: Supercharge Your Projects”

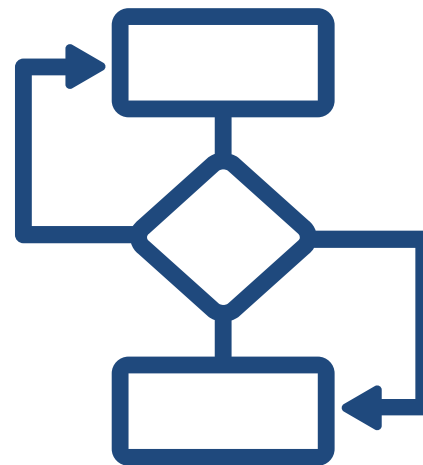
Agenda

- Overview MATLAB and Python Integration
- Data Exchange
- Co-execution
 - ❑ Call Python from MATLAB
 - ❑ Call MATLAB from Python
- Deployment
- Q&A

Overview MATLAB and Python Integration



Data Exchange



Co-Execution



Deployment

Data Exchange between MATLAB and Python

Scalar values only



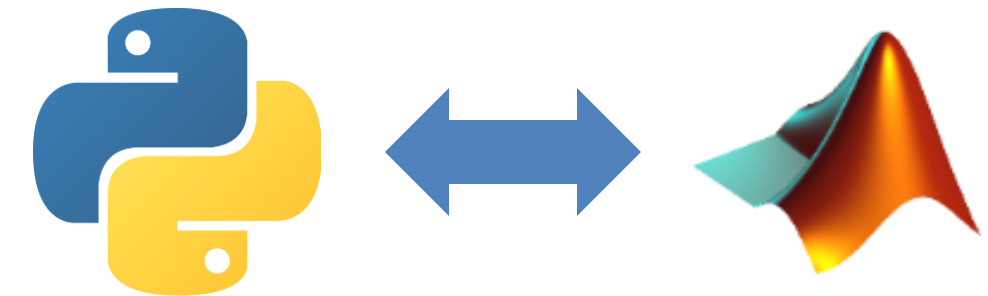
Data Exchange

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

Python Input Argument Type	Resulting MATLAB Data Type
float	double
complex	complex double
int	int64
long (Python 2.7 only)	int64
float(nan)	NaN
float(inf)	Inf
bool	logical
str	char
unicode (Python 2.7 only)	char
dict	struct if all keys are strings not supported otherwise



Create List, Tuple, and Dictionary Types

- This table shows the commands for creating **list**, **tuple**, **dict** types.*

Python list – []	MATLAB py.list
>>> ['Kantika', 'Wongkasem', 'Fah']	>> py.list({'Kantika','Wongkasem','Joseph'})
>>> [[1,2],[3,4]]	>> py.list({py.list([1,2]),py.list([3,4])})
Python tuple – ()	MATLAB py.tuple
>>> ('Kantika', 27, 'Engineer')	>> py.tuple({'Kantika',19,'Engineer'})
Python dict – {}	MATLAB py.dict
>>> {'Ink': 424 , 'Pink': 543 , 'Blue': 320}	>>py.dict(pyargs('Ink',424,'Pink',543,'Blue',320))
	For information about passing keyword arguments, see pyargs

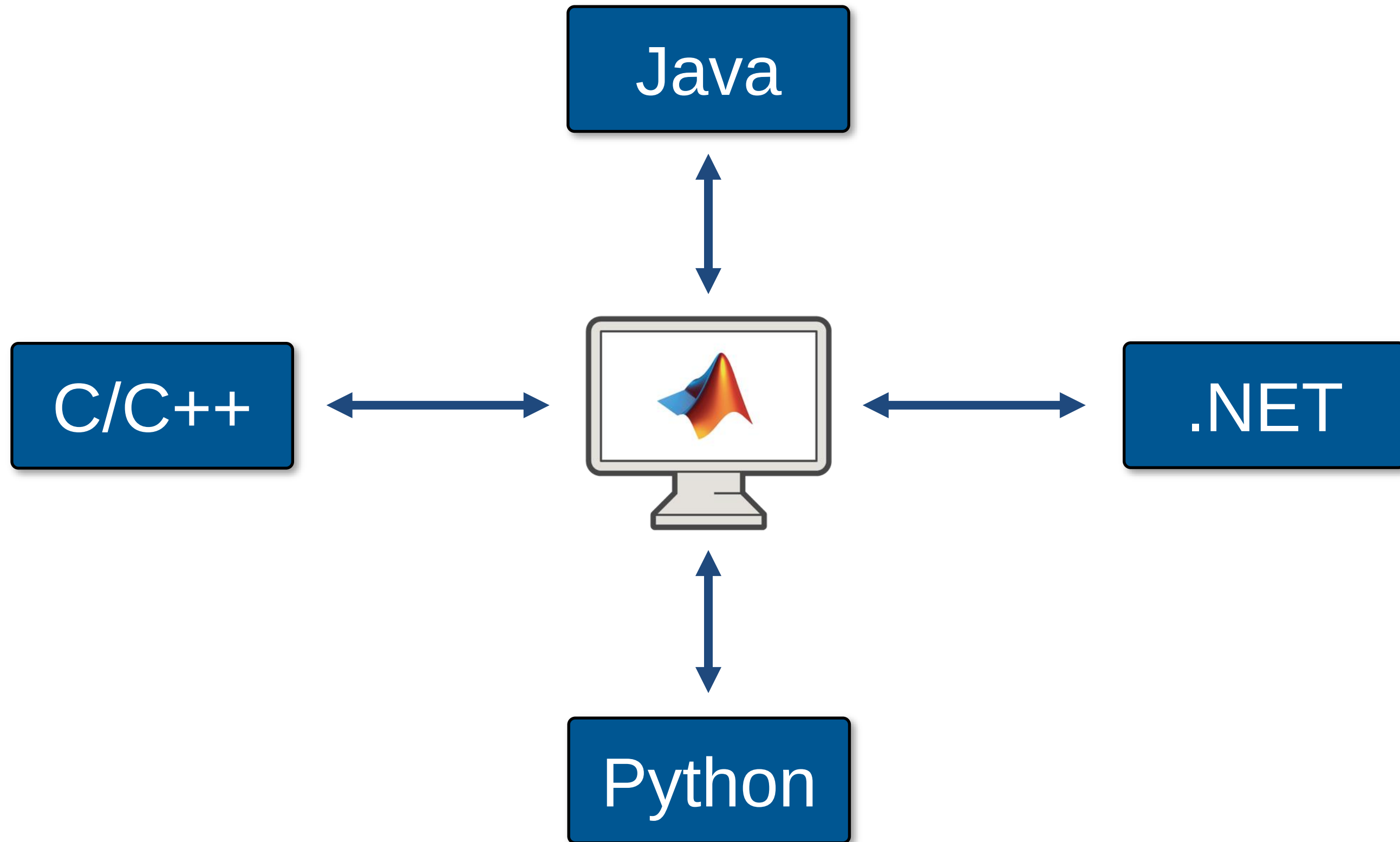
Data Exchange

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

MATLAB provides flexible integration with multiple languages



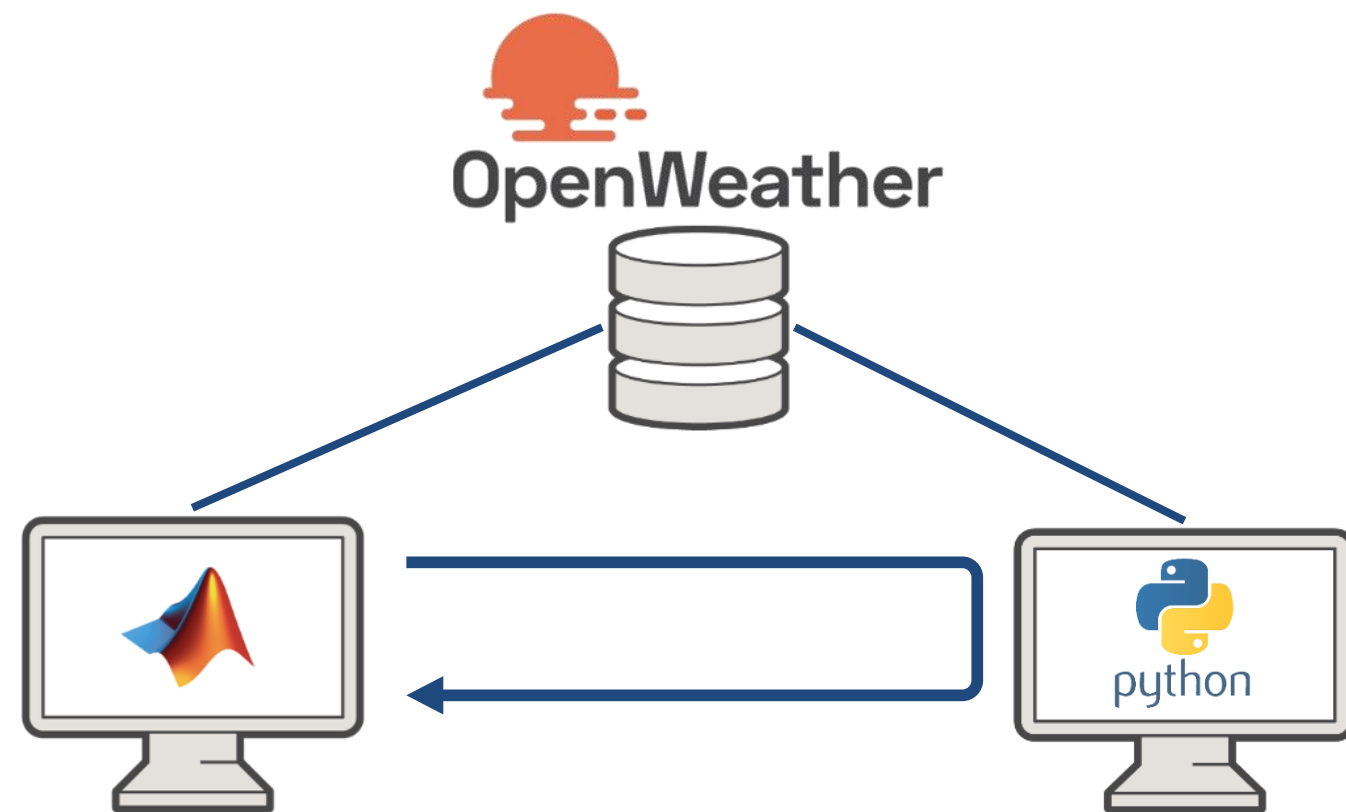
Data Exchange

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

Example: Weather Prediction



```
1 % Check that MATLAB can recognize the Python installation. Use pyversion if you are using a MATLAB version prior to R2019b.
2 % pyversion()
3 % pyenv

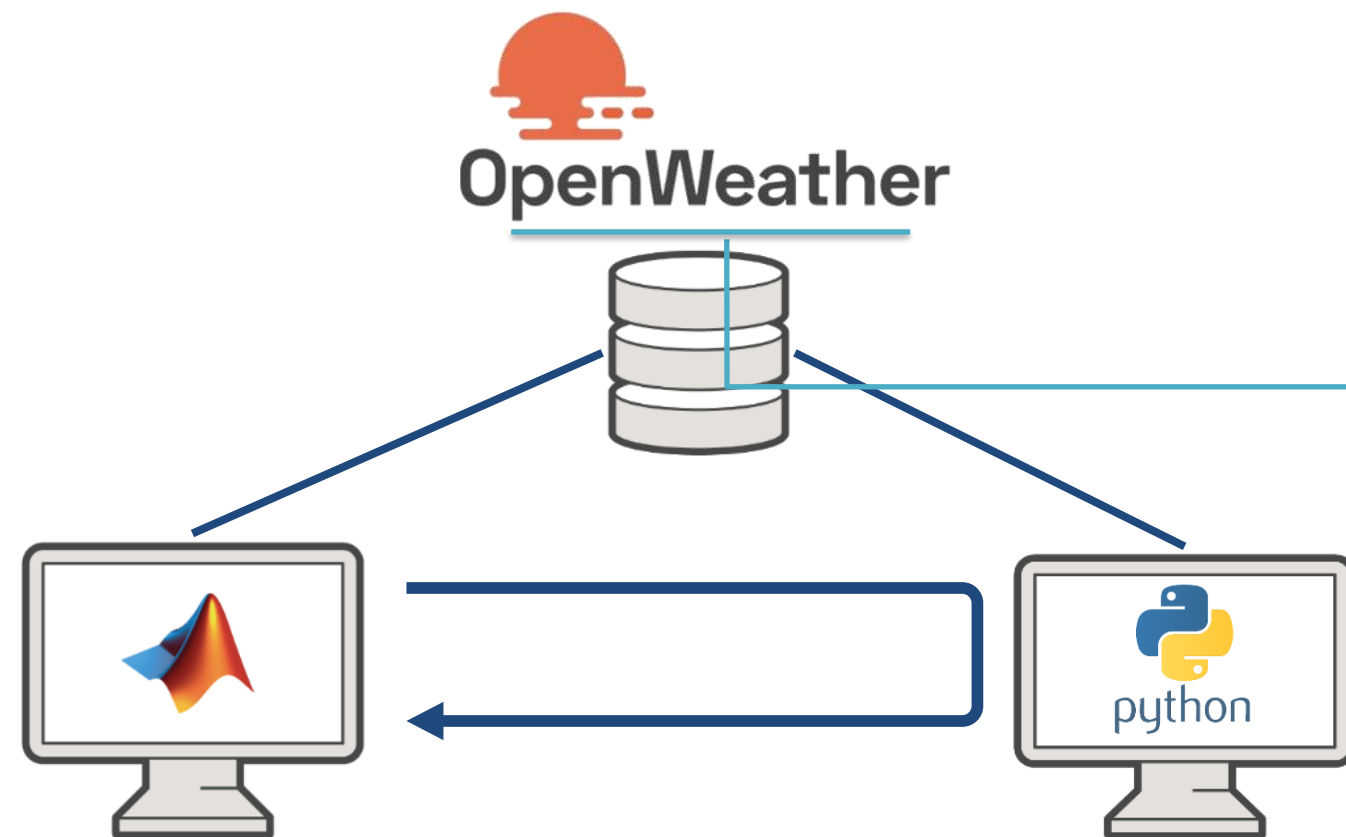
4 % Test a simple function.
5 % py.math.sqrt(42)

6 % Pass arguments using pyargs.
7 % textwrap.wrap("Hello World",width=2)

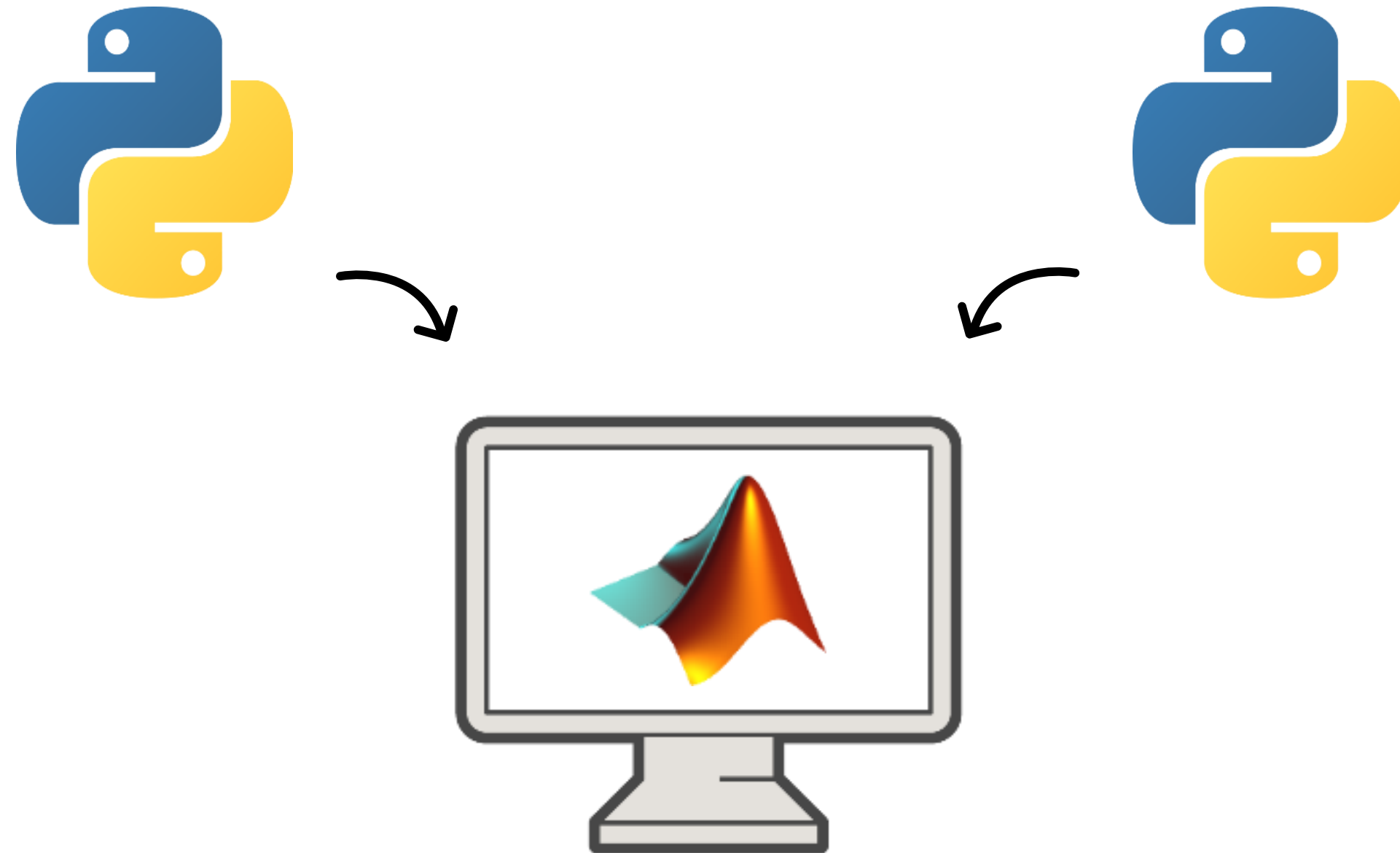
8 % py.textwrap.wrap("Hello world!",pyargs('width',2)) % errors
9 % py.textwrap.wrap("Hello world!",pyargs('width',int8(2)))

10 % Read weather data using Python module
11 % OpenWeather global services
```


Data Access



```
def get_current_weather(city, country, apikey):  
    # get current conditions in specified location  
    # get_current_weather('boston','us',key)  
    import urllib.request  
    import json  
    # read current conditions  
    try:  
        url = "https://api.openweathermap.org/data/2.5/weather?q="+city+", "+country+"&appid="+apikey  
        response = urllib.request.urlopen(url)  
        html = response.read()  
        json_data = json.loads(html)  
  
    except urllib.error.URLError:  
        # if weather API doesnt work, read the file  
        json_data = read_backup(city)  
  
    return json_data
```

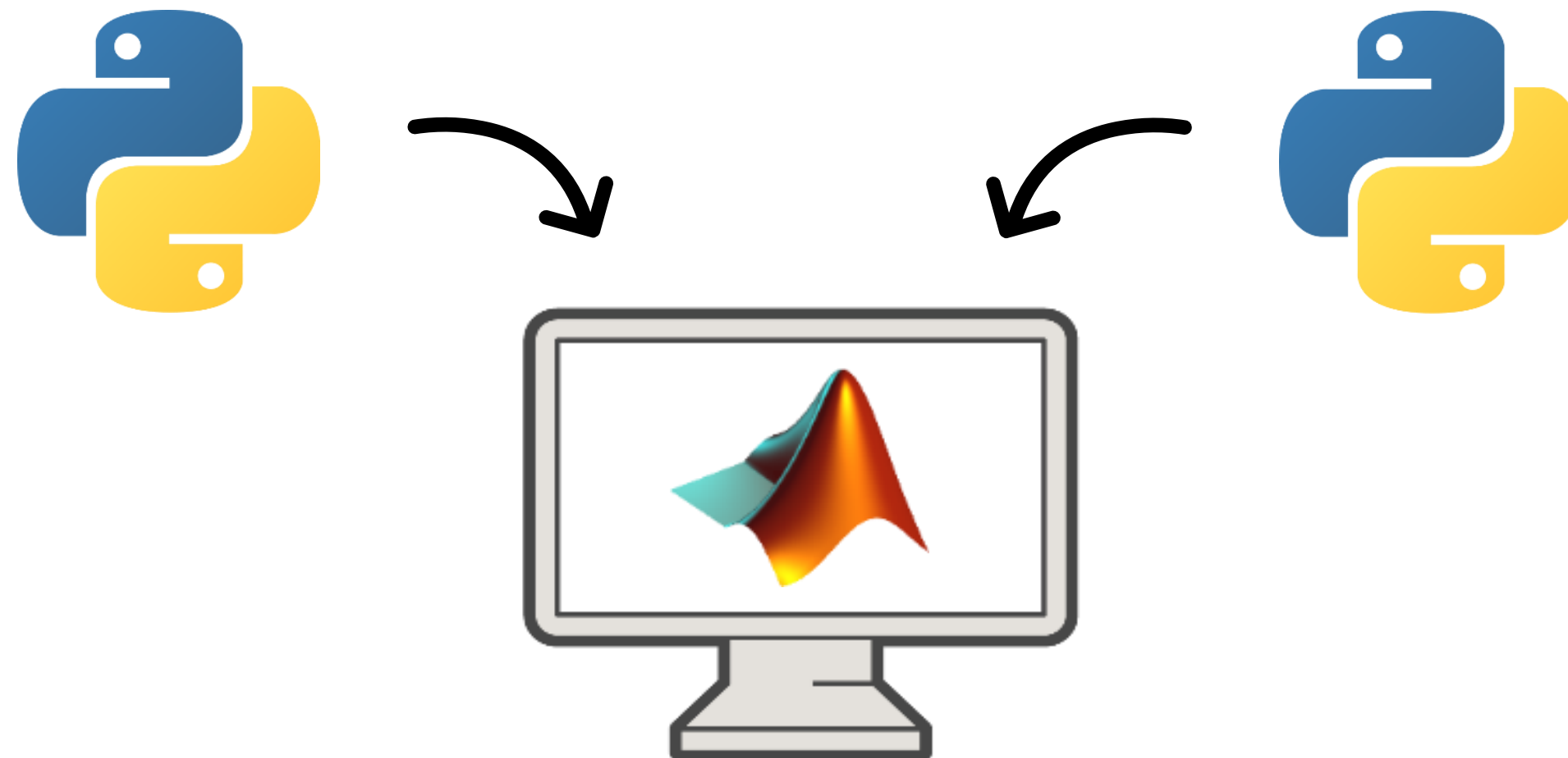



Call Python from MATLAB

Why Call Python from MATLAB?

Already working in MATLAB, and:

- Want to reuse existing Python code
- Need functionality that is only available in Python



Data Exchange

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

Setup tips



- Use `pyenv` to update settings (`pyversion` before R2019b)
- Ensure all code is on path
- Check environment settings, depending on how you set up python



- How to select the MATLAB engine version if I have multiple MATLAB releases to interface?

Setup

Ensure Python is installed and confirm the version with `pyenv`.

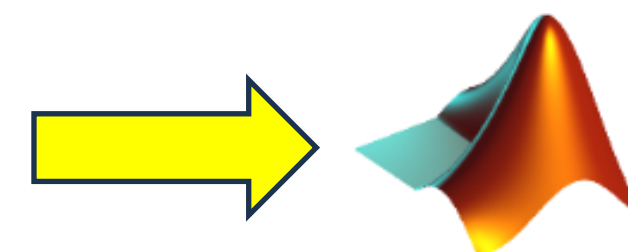
We will provide the path to the correct executable for MATLAB Online.

```
pyenv
```

```
ans =
```

```
PythonEnvironment with properties:
```

```
Version: "3.9"  
Executable: "C:\Users\KantikaWongkasem\anaconda3\python.EXE"  
Library: "C:\Users\KantikaWongkasem\anaconda3\python39.dll"  
Home: "C:\Users\KantikaWongkasem\anaconda3"  
Status: Loaded  
ExecutionMode: OutOfProcess  
ProcessID: "4668"  
ProcessName: "MATLABPyHost"
```



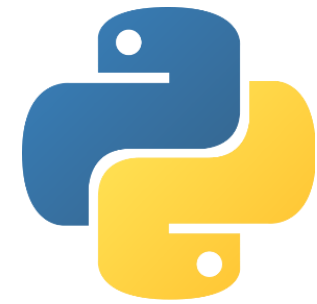
Data Exchange

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

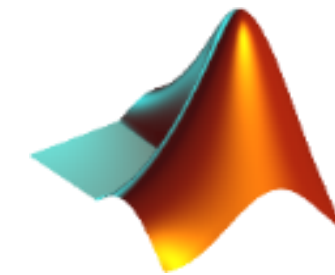
Deployment

Syntax differences when calling Python from MATLAB



Python

```
>>> import math  
>>> math.sqrt(42)
```



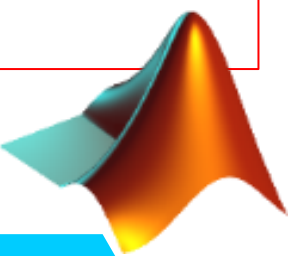
MATLAB

```
>> py.math.sqrt(42)
```

```
>>> print('hello', 'world', sep=', ')
```



```
>> py.print('hello', 'world', ...  
           pyargs('sep', ', '))
```



Data Exchange

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

Calling User-Defined Python libraries from MATLAB

Data Exchange

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

Use the weather.py module to get the air quality for Paris. This is a user-defined Python module which includes functions to read and parse the current and forecasted weather data by location.

```
jsonData = py.weather.get_current_weather("Paris","France",apikey.Key)
```

```
jsonData =  
    Python dict with no properties.
```

```
    {'coord': {'lon': 2.35, 'lat': 48.85}, 'weather': [{'id': 803, 'main': 'Cloudy'}
```

Parse the json data returned from the weather API.

The Python dictionary can be represented as a MATLAB struct.

```
weatherData = py.weather.parse_json(jsondata);  
struct(weatherData)
```

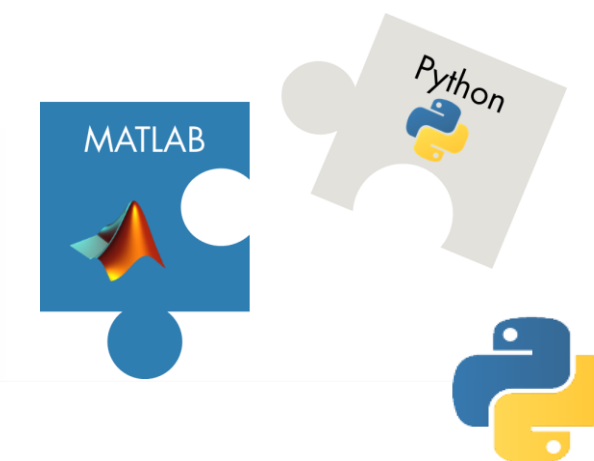
```
ans = struct with fields:  
    temp: 18.7100  
    feels_like: 17.3000  
    temp_min: 17.7800  
    temp_max: [1x1 py.int]
```

Use a function (prepData.m) to prepare data for machine learning (create a table with the expected variable names, preprocessing steps, etc).

```
currentData = prepData(weatherData)
```

```
currentData = 1x12 table
```

	DateLocal	city	StateName	T	P	DP	RH	WindDir	WindSpd		
1	01-Jul-2020 11:...	"Paris"	Ile de France	21.6200	20.2600	349.2200	1010	5.1000		73	



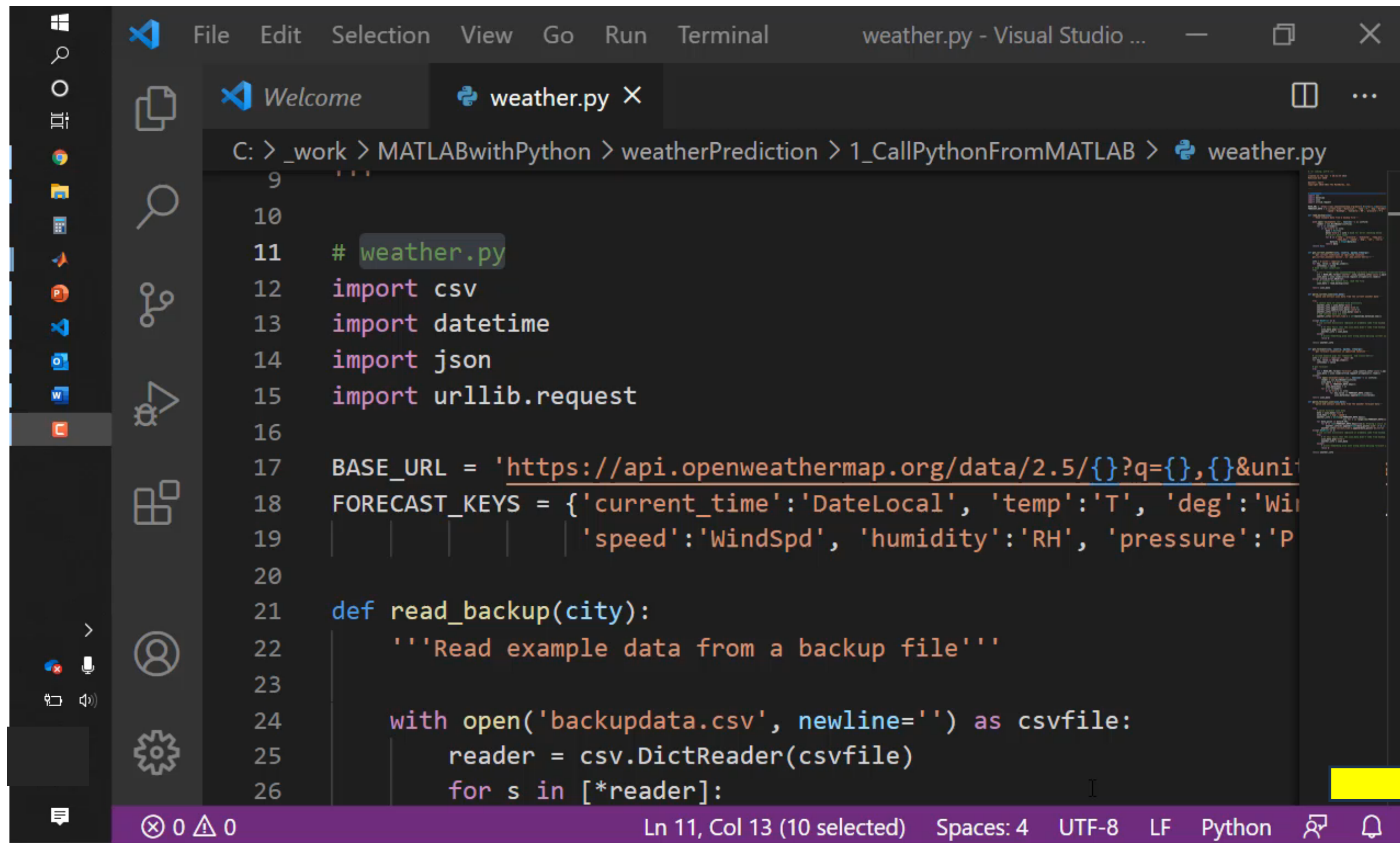
Calling User-Defined Python libraries from MATLAB

Data Exchange

Co-Execution

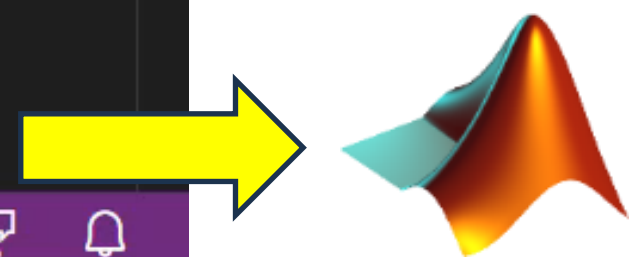
- Call Python from MATLAB
- Call MATLAB from Python

Deployment



```
File Edit Selection View Go Run Terminal weather.py - Visual Studio ...
Welcome weather.py X
C: > _work > MATLABwithPython > weatherPrediction > 1_CallPythonFromMATLAB > weather.py
9
10
11 # weather.py
12 import csv
13 import datetime
14 import json
15 import urllib.request
16
17 BASE_URL = 'https://api.openweathermap.org/data/2.5/{ }?q={ },{ }&uni
18 FORECAST_KEYS = {'current_time': 'DateLocal', 'temp': 'T', 'deg': 'Wi
19 | | | | | 'speed': 'WindSpd', 'humidity': 'RH', 'pressure': 'P
20
21 def read_backup(city):
22     '''Read example data from a backup file'''
23
24     with open('backupdata.csv', newline='') as csvfile:
25         reader = csv.DictReader(csvfile)
26         for s in [*reader]:
```

Ln 11, Col 13 (10 selected) Spaces: 4 UTF-8 LF Python



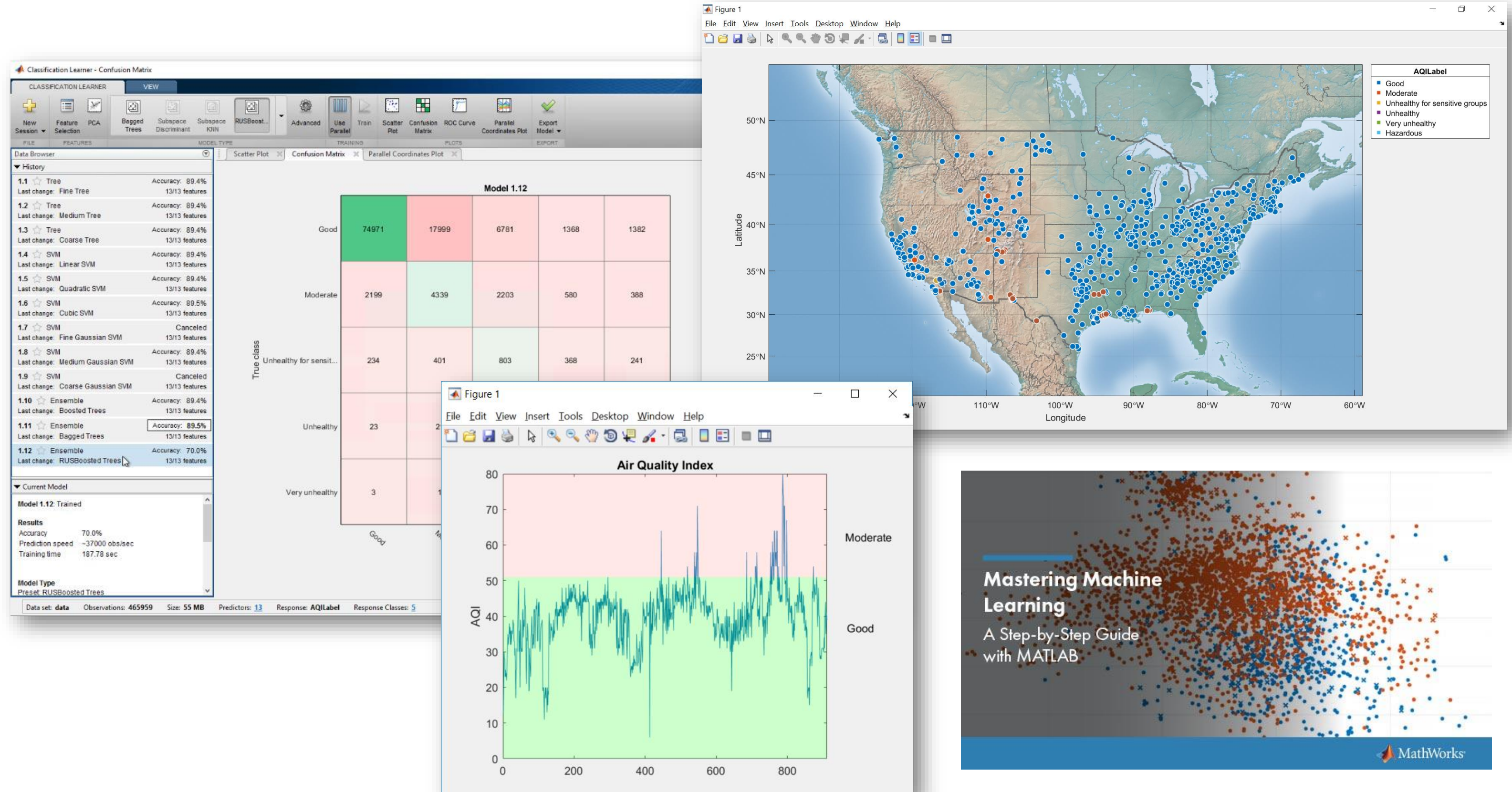
Air Quality Prediction Model

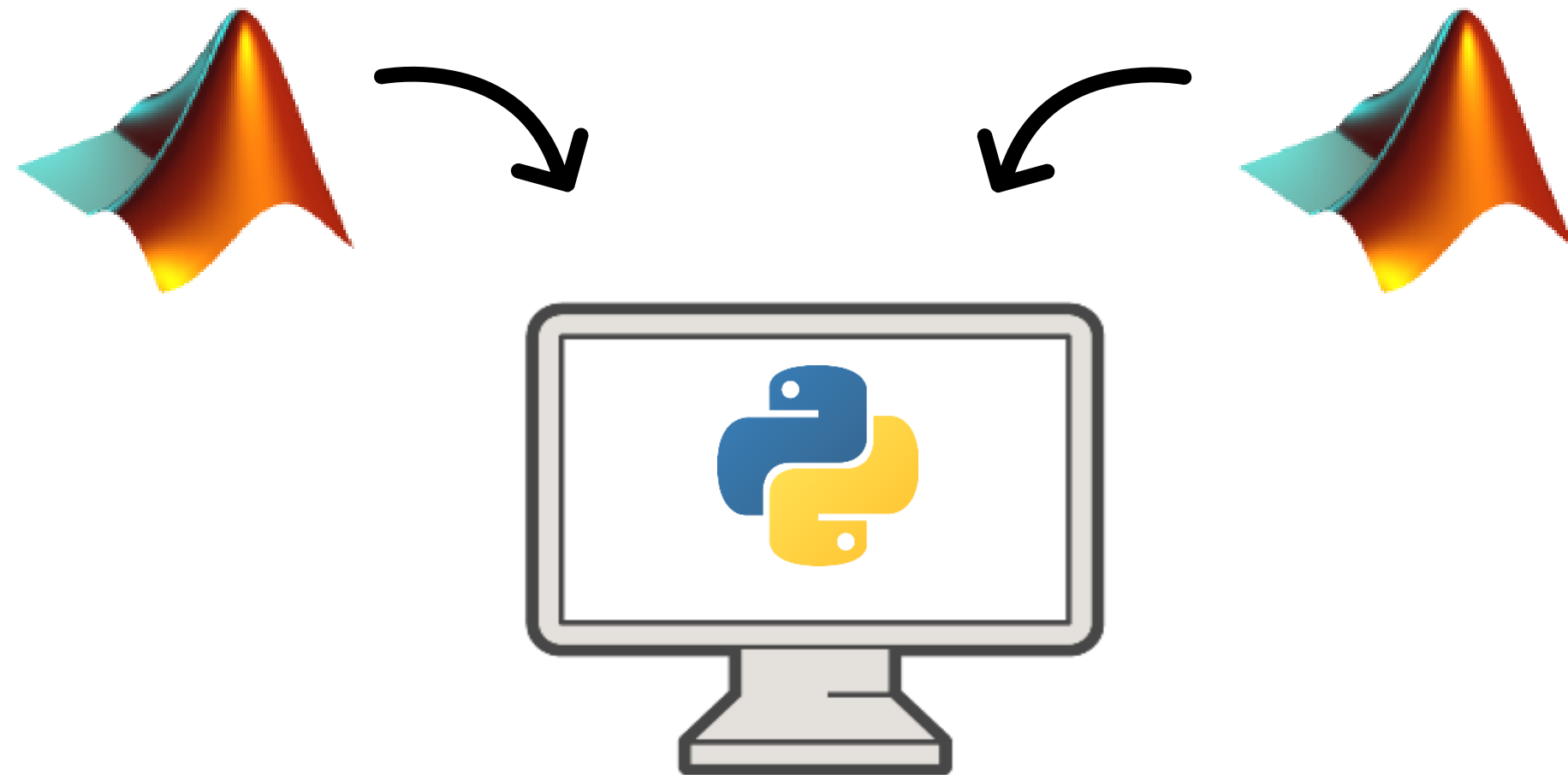
Data Exchange

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment



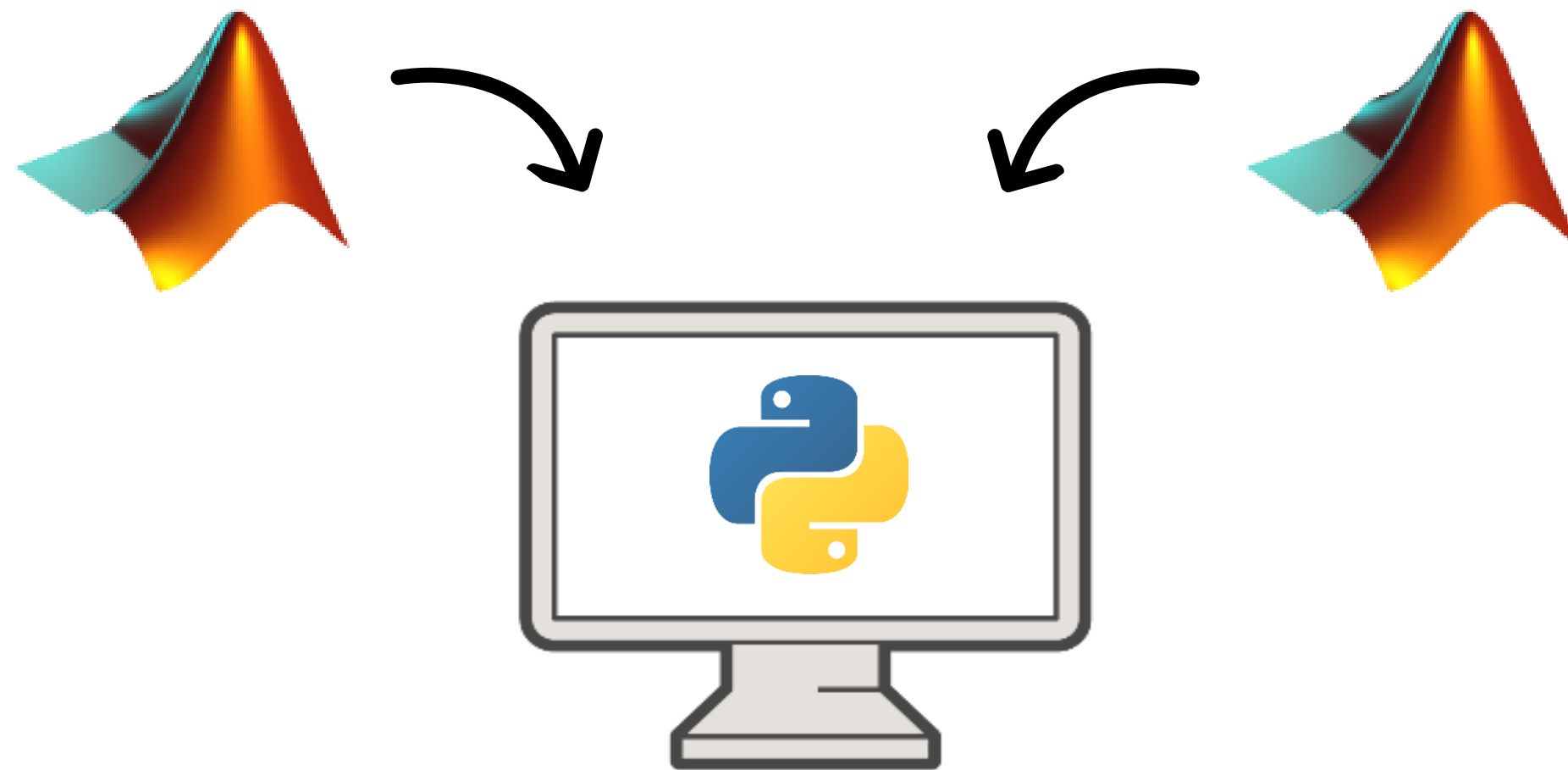


Call MATLAB from Python

Why call MATLAB from Python?

Already working in Python, and:

- Want to reuse existing MATLAB code
- Need functionality available in MATLAB
- Want to collaborate with MATLAB users



Data Exchange

Co-Execution

- Call Python from MATLAB

- Call MATLAB from Python

Deployment

Install MATLAB Engine API for Python

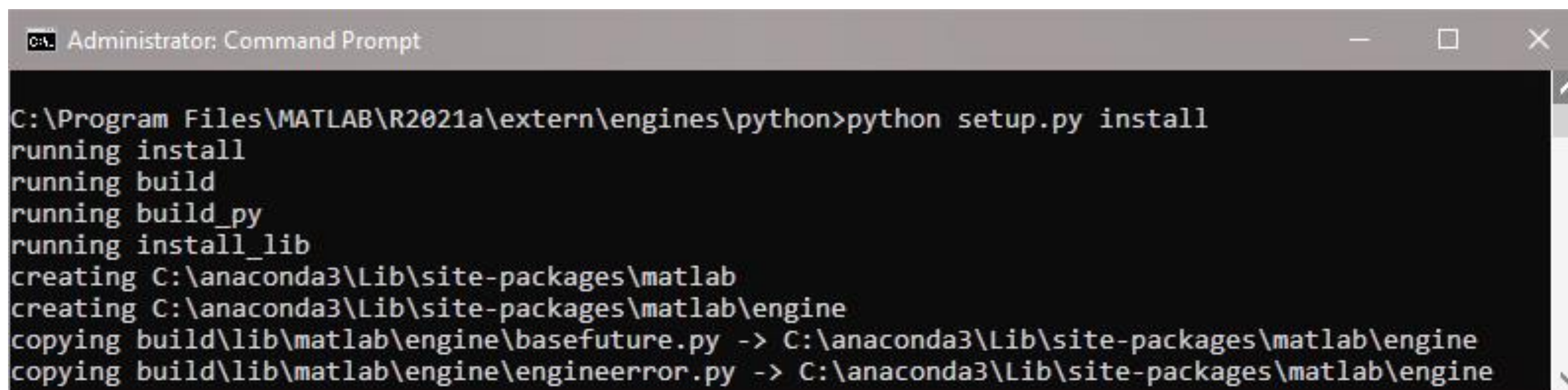
- Linux or macOS

```
cd "matlabroot/extern/engines/python"  
python setup.py install
```

- Windows

```
cd "matlabroot\extern\engines\python"  
python setup.py install
```

matlabroot - full path to the folder where MATLAB is installed



```
Administrator: Command Prompt  
  
C:\Program Files\MATLAB\R2021a\extern\engines\python>python setup.py install  
running install  
running build  
running build_py  
running install_lib  
creating C:\anaconda3\Lib\site-packages\matlab  
creating C:\anaconda3\Lib\site-packages\matlab\engine  
copying build\lib\matlab\engine\basefuture.py -> C:\anaconda3\Lib\site-packages\matlab\engine  
copying build\lib\matlab\engine\engineerror.py -> C:\anaconda3\Lib\site-packages\matlab\engine
```

Data Exchange

Co-Execution

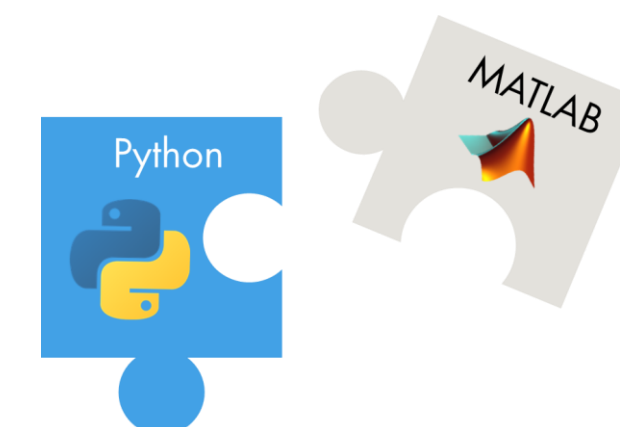
- Call Python from MATLAB

- Call MATLAB from Python

Deployment

Call MATLAB from Python

MATLAB Engine API



- Install MATLAB Engine API for Python

```
$ cd "C:\Program Files\MATLAB\R2021b\extern\engines\python"
```

```
$ python setup.py install
```

- Start a MATLAB process

```
>>> import matlab.engine
```

```
>>> m = matlab.engine.start_matlab()
```

- Call MATLAB functions

```
>>> x = m.sqrt(float(42))
```

Data Exchange

Co-Execution

- Call Python from MATLAB

- Call MATLAB from Python

Deployment

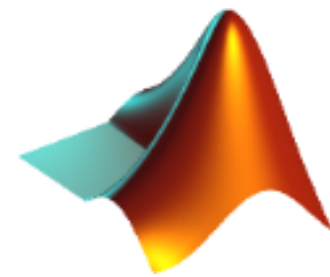
Syntax differences when calling MATLAB from Python

Data Exchange

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

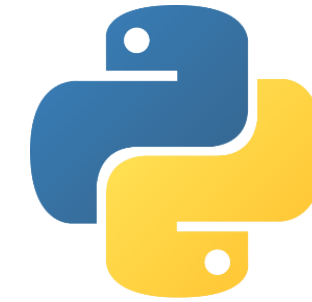


MATLAB

```
>> [s,sidx] = sort(x)
```

```
>> foo(x)
```

```
>> C = A + B
```



Python

```
m = matlab.engine.start_matlab()
```

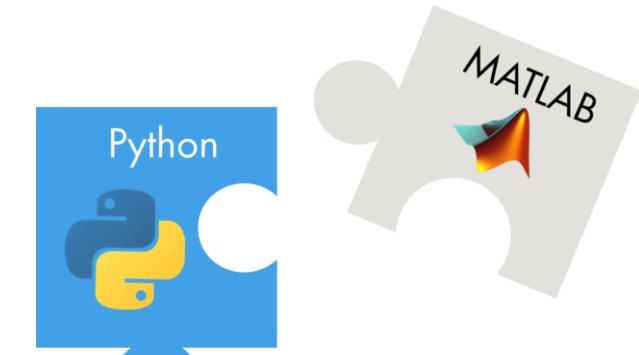
```
>>> s = m.sort(x,nargout=2)
```

```
>>> m.foo(x,nargout=0)
```

```
>>> C = m.plus(A,B)
```


Integrate the MATLAB model in a Python environment

[CallMATLABfromPython.ipynb](#)



Data Exchange

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

Predict Air Quality: Calling MATLAB from Python Using MATLAB Engine API

Import the weather data and use the MATLAB code to predict air quality.

This example uses the MATLAB Engine API, which runs a MATLAB session. More info and set up instructions are included here: <https://www.mathworks.com/help/matlab/matlab-engine-for-python>

Modeling & Training

Machine Learning

Data Exchange

Data Preparation

Modeling
& Training

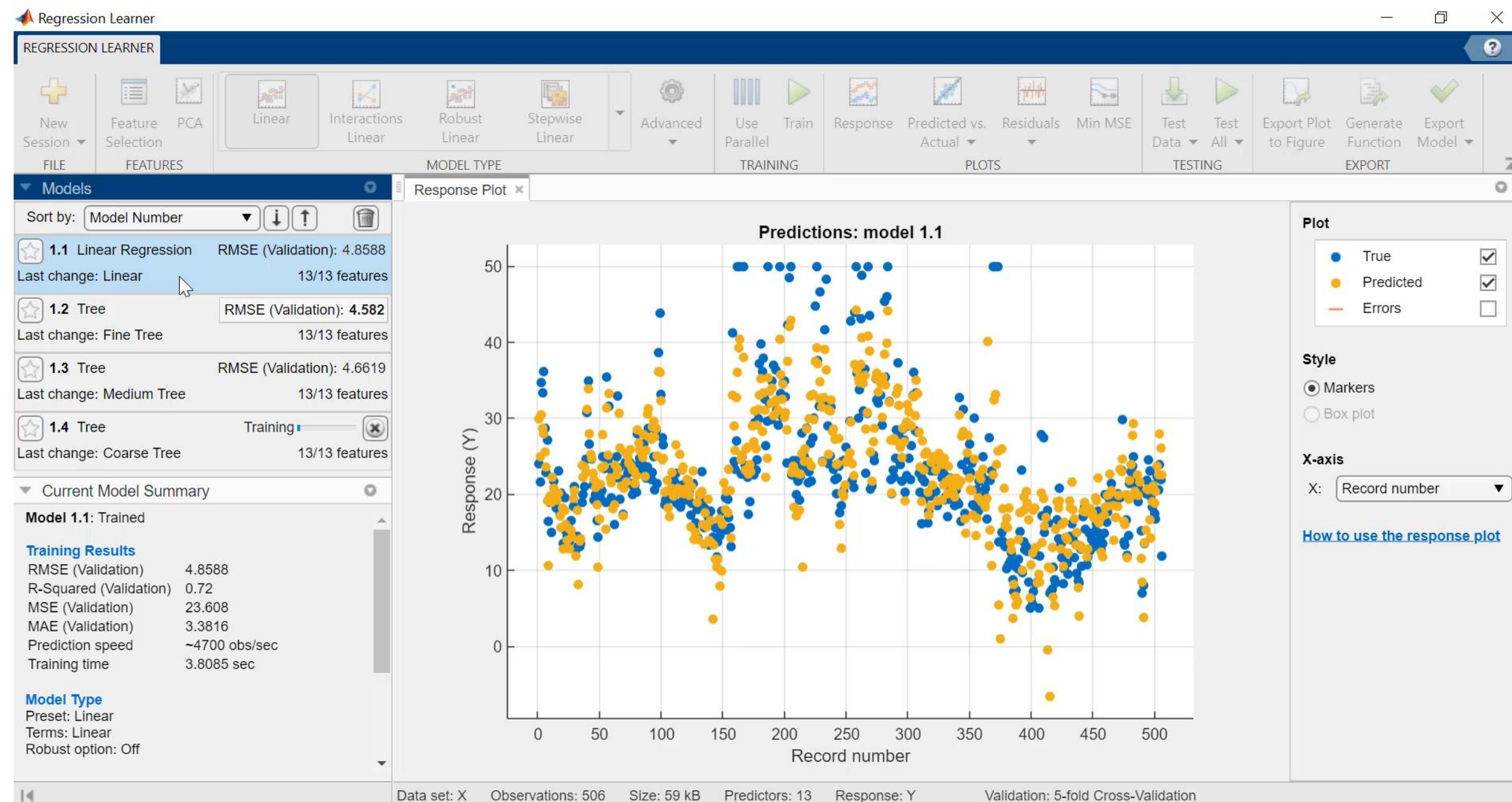
Simulation & Test

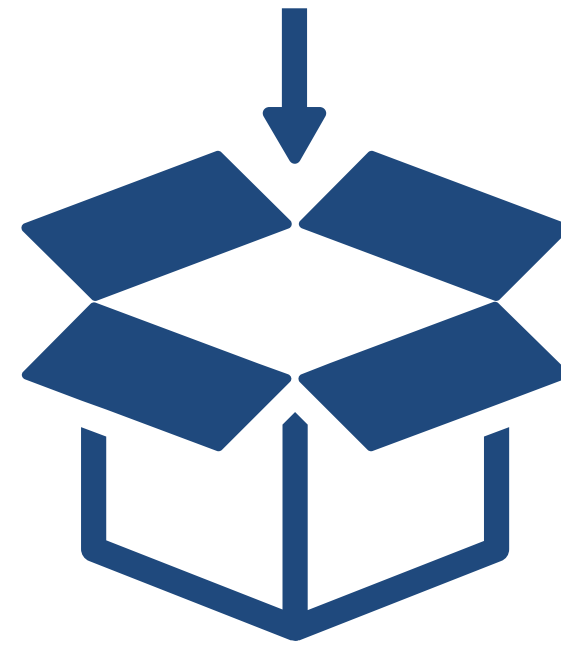
Deployment

Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

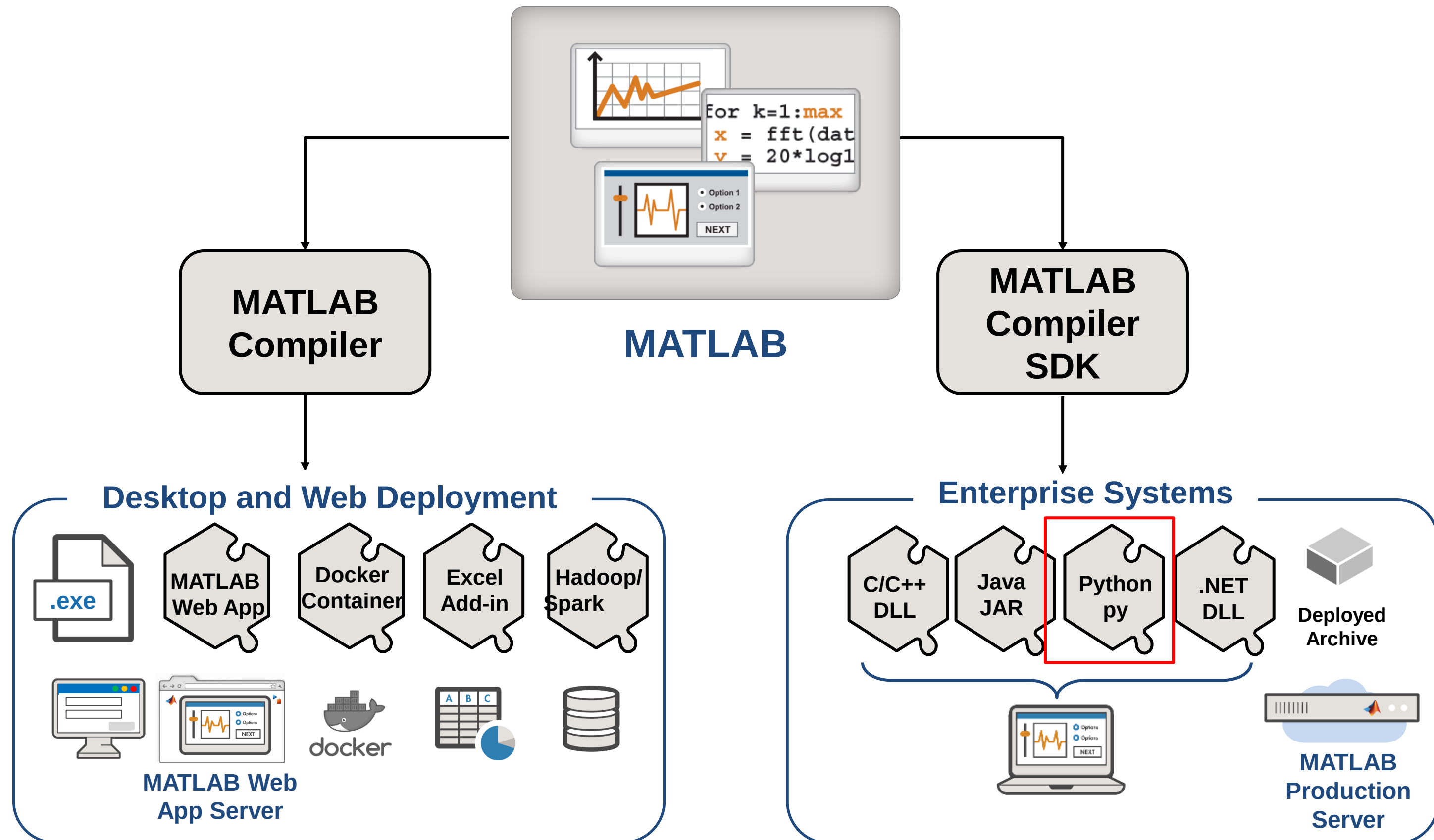




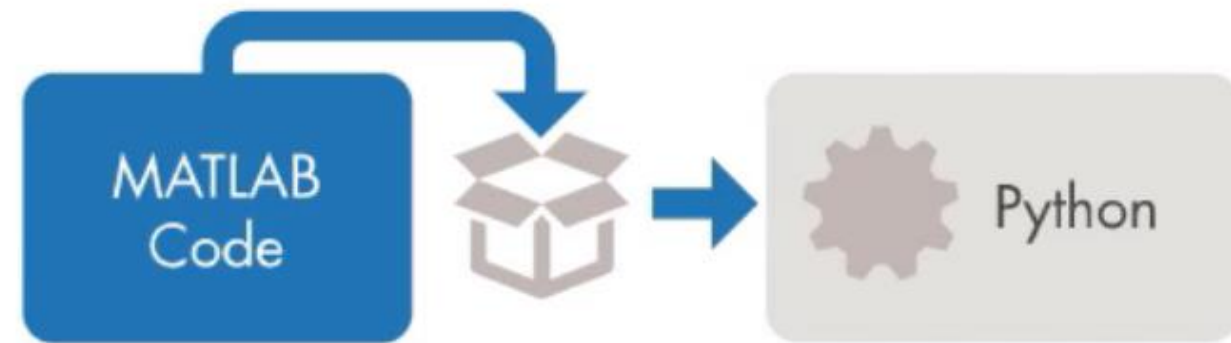
Deployment

Generate Python library

Deployment of MATLAB Programs



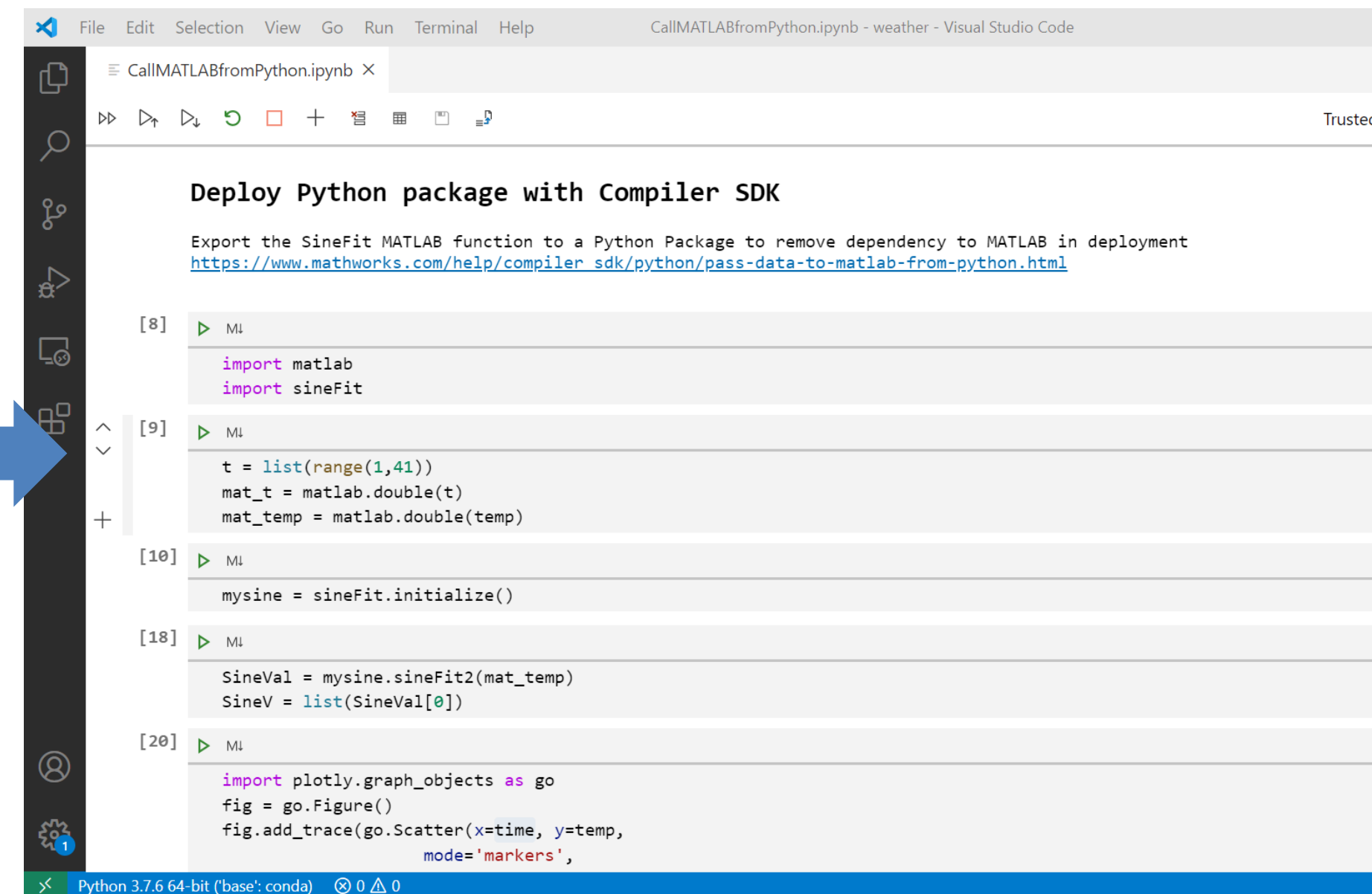
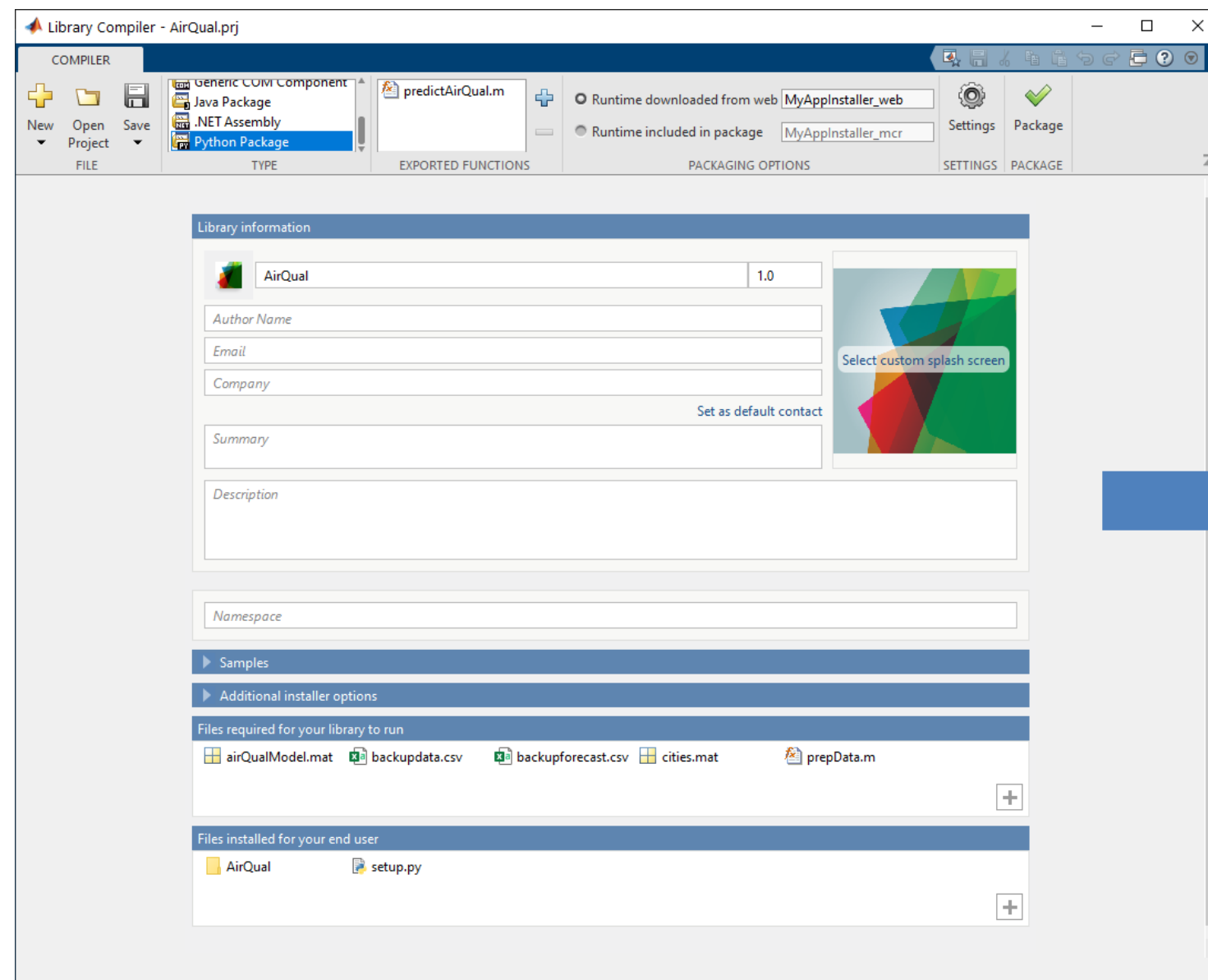
Generate Python library from MATLAB functions



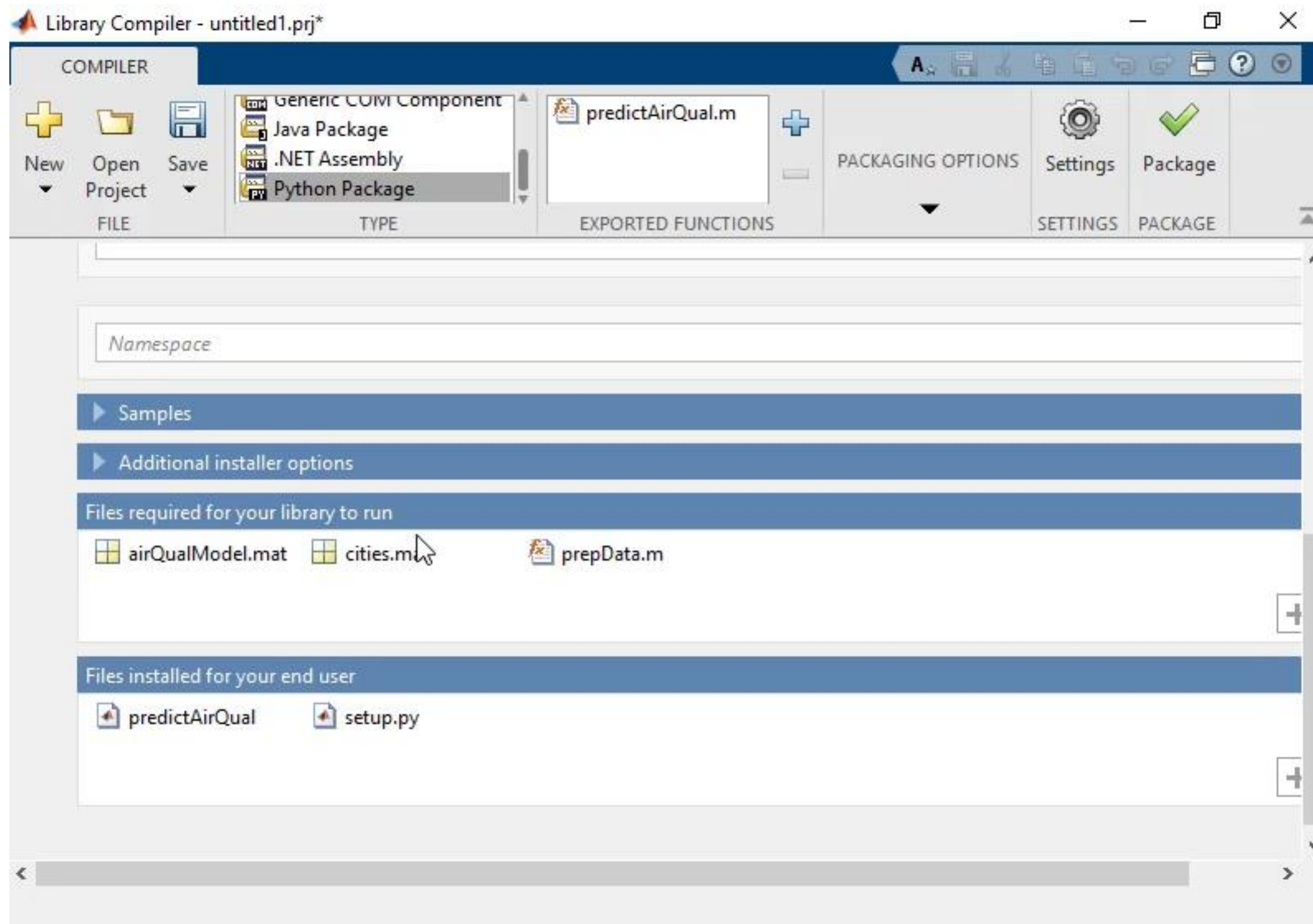
Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment



Generate Python library from MATLAB functions

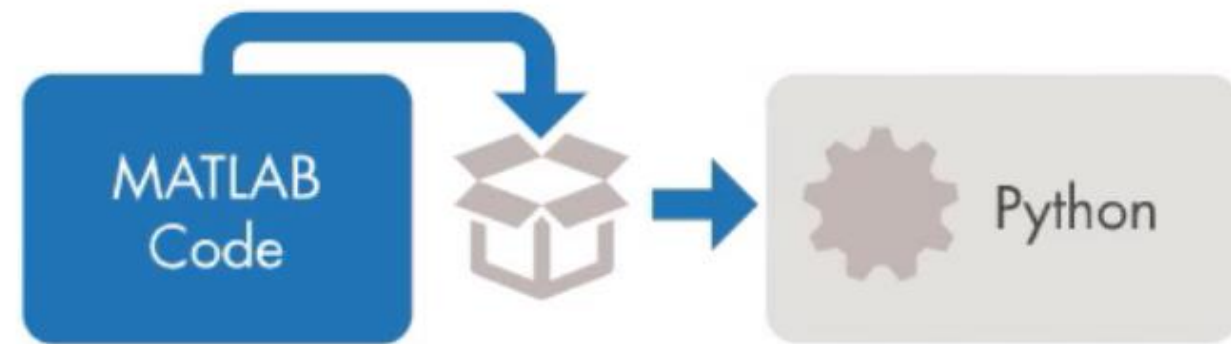


Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

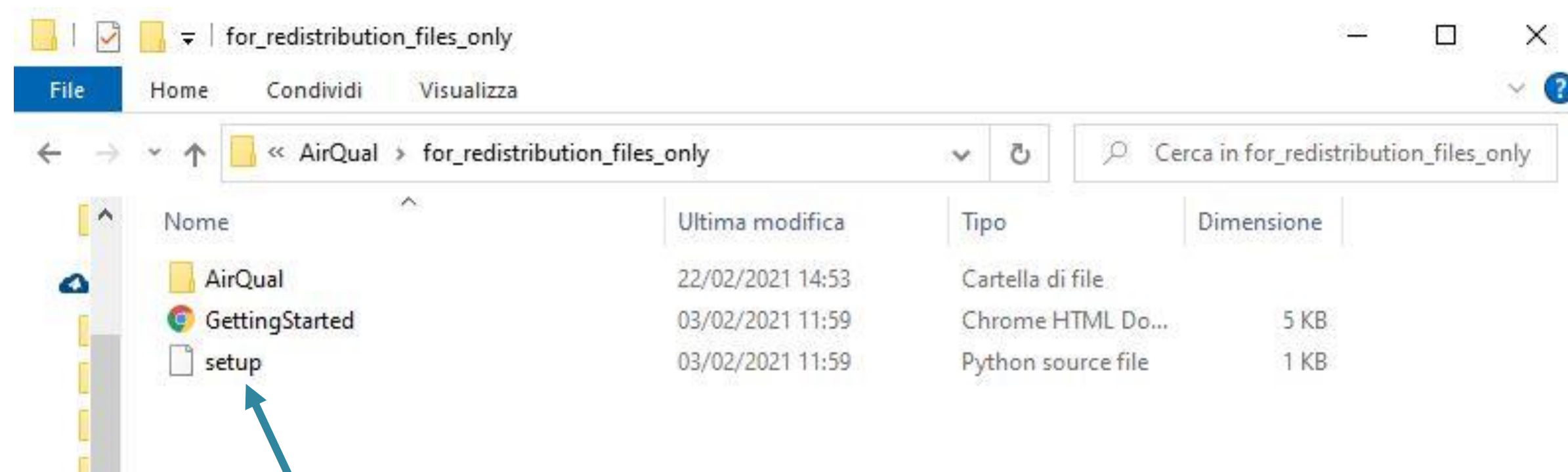
Generate Python library from MATLAB functions



Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

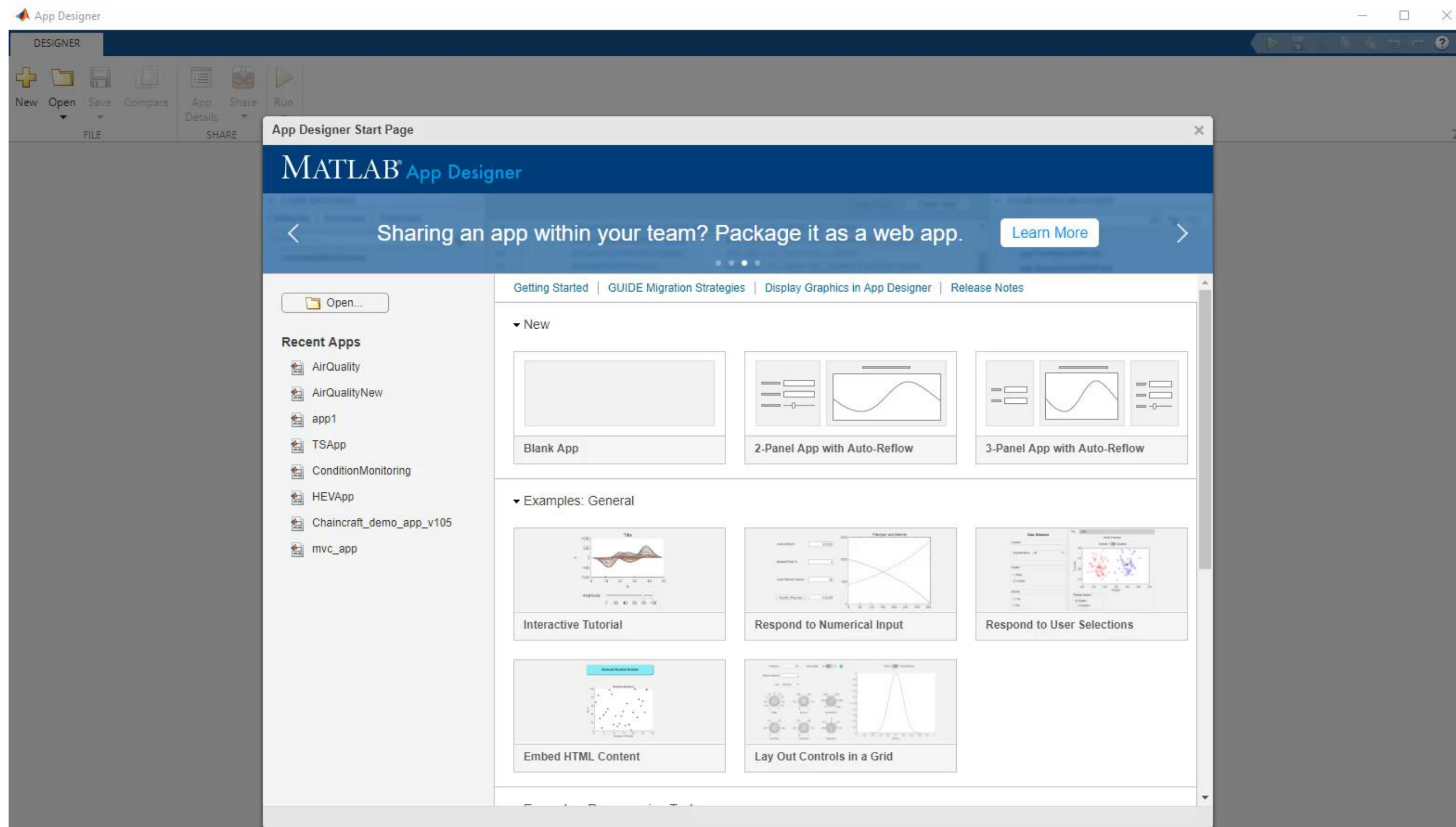
Deployment



Install library

MATLAB App Designer

App development for Non-Programmers



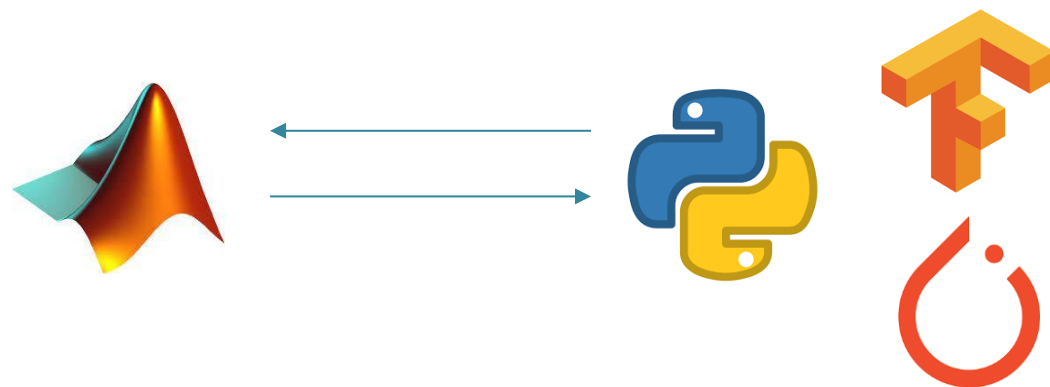
Co-Execution

- Call Python from MATLAB
- Call MATLAB from Python

Deployment

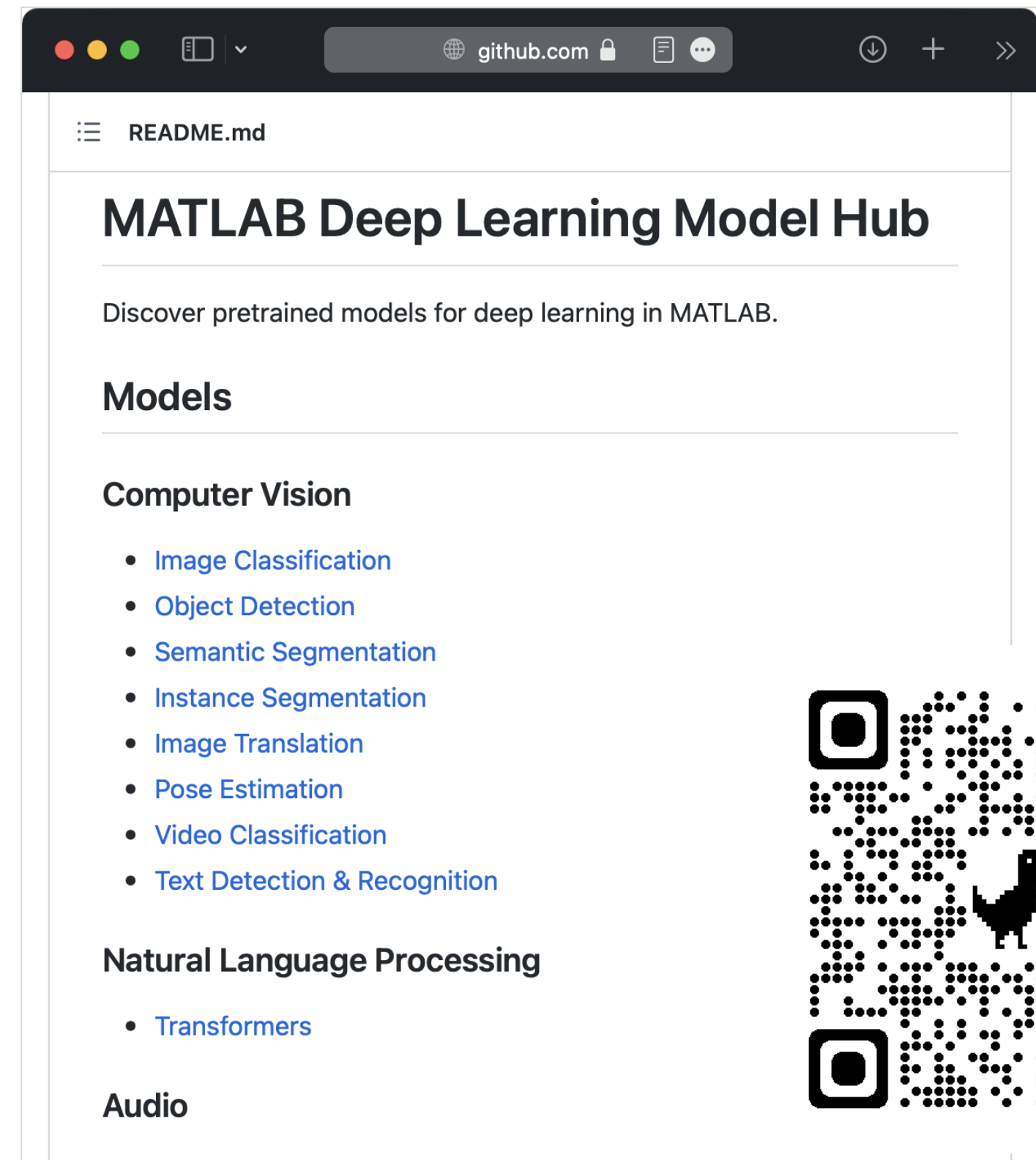
Deep Learning

Co-execution



Model import/export

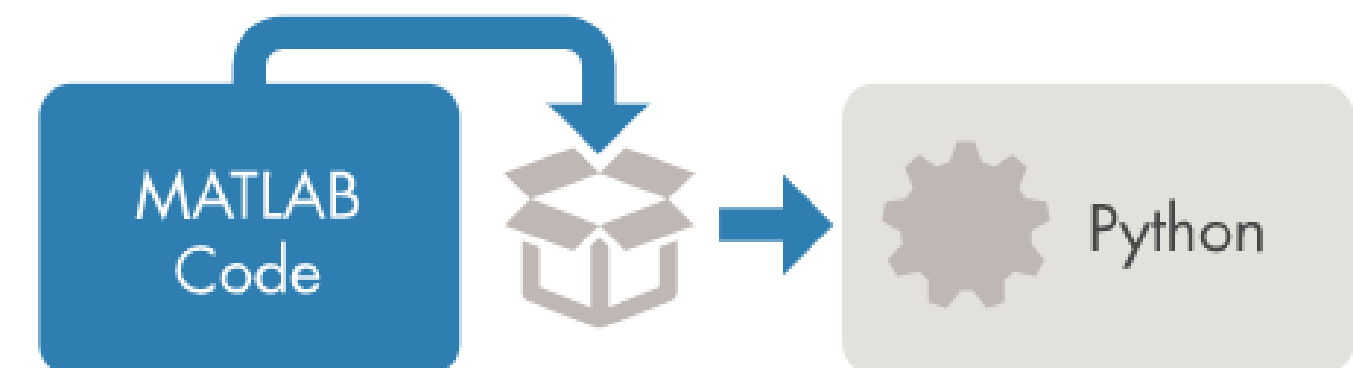
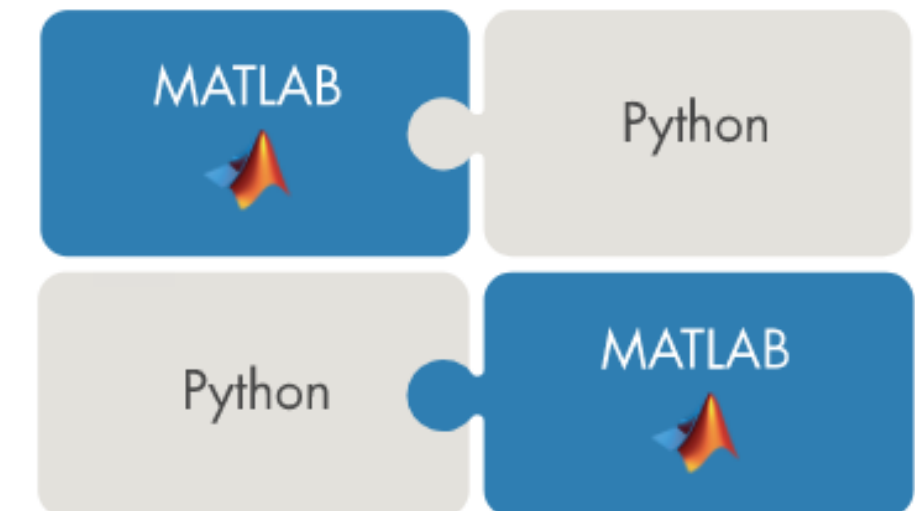
```
>> importTensorFlowNetwork("SavedModel")  
>> importONNXNetwork("model.onnx")  
>> exportONNXNetwork(net, "model.onnx")
```



<https://github.com/matlab-deep-learning/MATLAB-Deep-Learning-Model-Hub>

Key Takeaways for MATLAB and Python Integration

- Access Data
 - Weather Data example `get_current_weather`
- Interoperability
 - Calling Python from MATLAB
 - Calling User-Defined Python libraries from MATLAB
 - Calling MATLAB from Python
- Deployment
 - Package Python library from MATLAB function
 - App Designer



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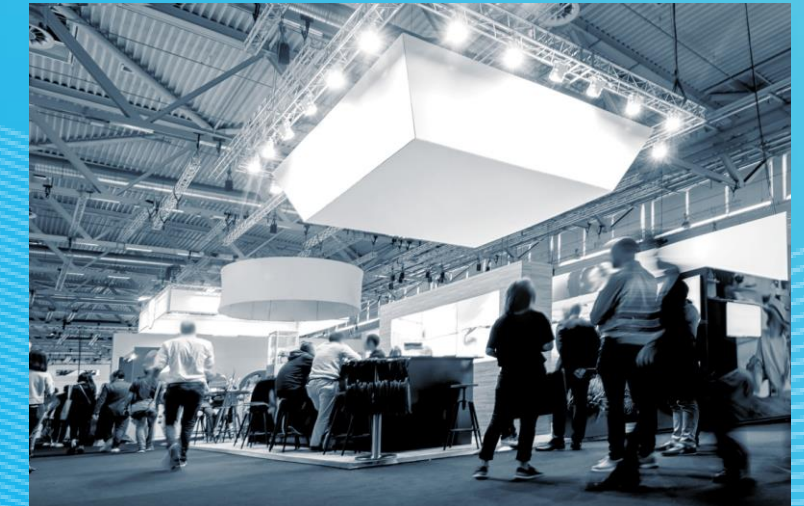
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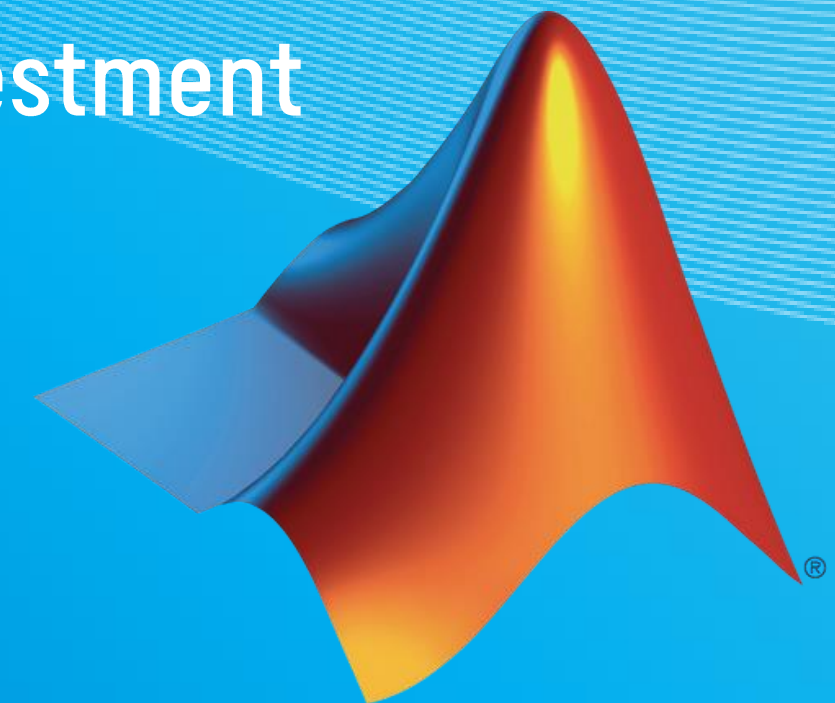
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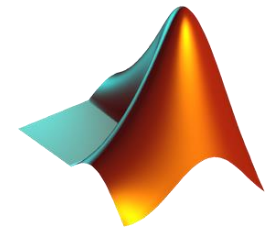


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